# Generative AI Can Harm Teaching

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University of Pennsylvania

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# Overview

1. In 14 Turkish schools, we **randomly assigned** 193 teachers to get a GPT-4o teaching assistant or teach as usual for one semester.
2. 78% used it, mostly to produce teaching material, with little back-and-forth.
3. Students' intrinsic motivation fell (–0.11 SD), most under teachers who already used AI heavily.
4. Confidence fell less, mainly under weaker teachers.
5. Standardized test scores did not move on average, but students of weaker teachers lost 0.13 SD.
6. Teachers' beliefs about AI did not change: a principal–agent problem.

# Teachers already use generative AI

60%

of U.S. K-12 teachers used an AI tool last school year

Gallup & Walton Family Foundation 2025

1/3

use it every week

Gallup & Walton Family Foundation 2025

No causal evidence on its effects on students.

# AI could increase or decrease the quality of teaching

## Optimistic

High-quality content.

Time freed for rehearsal, relationships and differentiation.

Oreopoulos et al. 2024

## Pessimistic

Teachers offload the pedagogical thinking.

Materials come out polished but generic.

Time saved might not be reinvested.

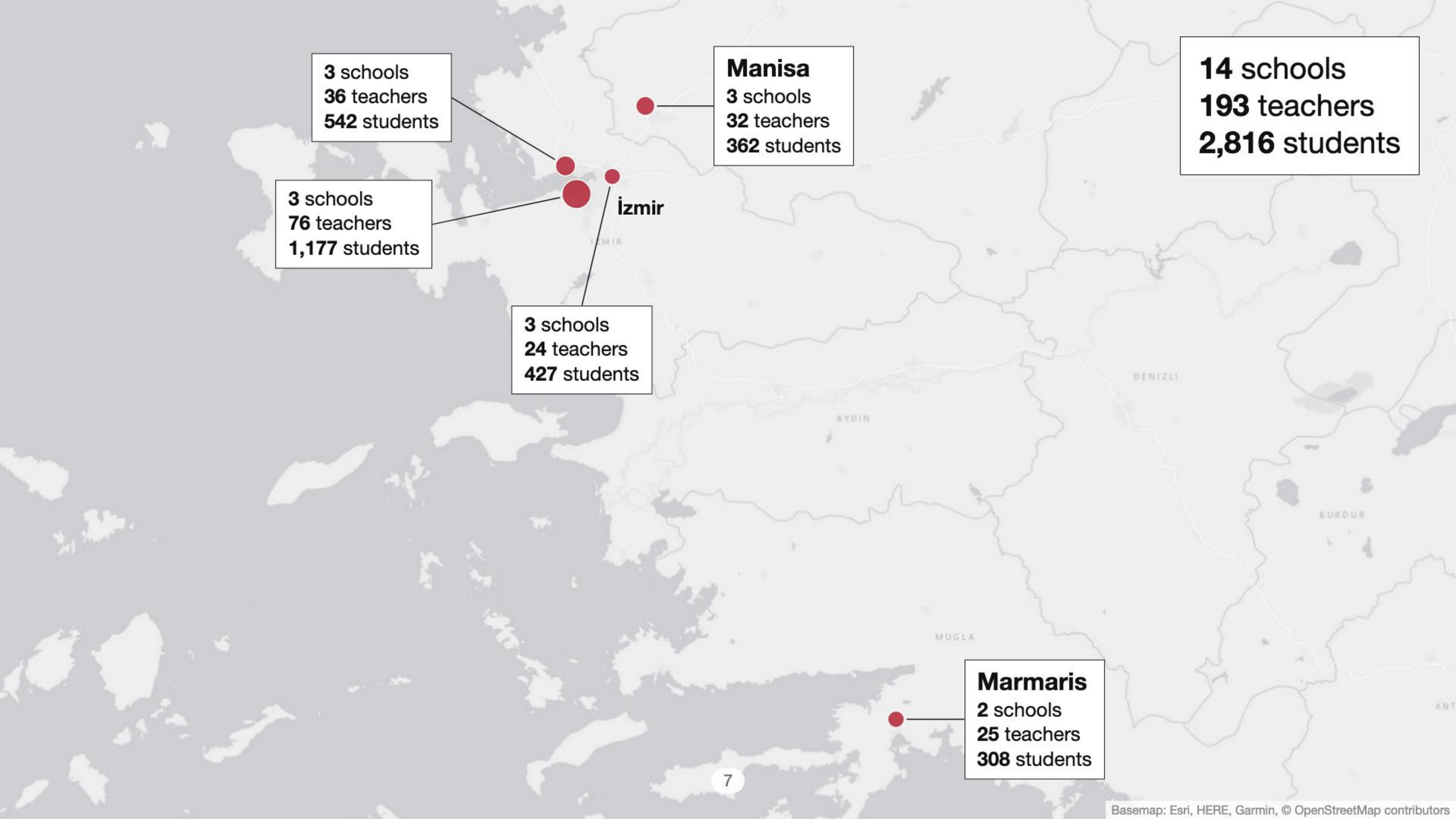
Doshi & Hauser 2024
Farber et al. 2014

# Five questions about what happens when teachers use AI

1. Adoption: do teachers use the tool?
2. Usage: what do they use it for?
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# A preregistered field experiment in 14 Turkish schools

| | |
|---|---|
| 14 | schools, grades 5–12 |
| 85 | school–department clusters |
| 193 | teachers |
| 2,816 | students |
| 14,198 | student–course observations |


3 schools
36 teachers
542 students
Manisa
3 schools
32 teachers
362 students
14 schools
193 teachers
2,816 students
3 schools
76 teachers
1,177 students
İzmir
3 schools
24 teachers
427 students
Marmaris
2 schools
25 teachers
308 students
İZMIR
AYDIN
DENIZLI
MUGLA
BURDUR
Basemap: Esri, HERE, Garmin, © OpenStreetMap contributors

## Control

Business-as-usual professional development

## AI Access

A one-hour training and a custom GPT-4o tool built on the national curriculum

## AI Access + Reminder

+ weekly emails showing each teacher's own usage, as a nudge

Both AI arms are **pooled** and compared with control.
Estimates are **intent-to-treat**.

Unit of randomization: the school–department



Details

ChatGPT-4 | Ders Plani Tasarlayici | Geri Bildirim Araci | Hata ve Kavram Yanılgısı Düzeltme | İdari İletisim Asistani | Ödev-Sınav Uretici

Öğrenci Destek ve Strateji Geliştirme Rehberi | Öğrenciler için stres yönetimi | Öğretmen İyi Oluş ve Destek Rehberi | Performans Analiz ve Geri Bildirim Asistani

Seviye ve İlgiye Göre Farklılaştırma | Unite Plani Tasarimi

Bu araç, Türkiye'deki öğretmenlere günlük ders planı hazırlarken rehberlik eder. Öğrencilerin ihtiyaçlarına uygun, anlamlı ve katılımcı ders etkinlikleri tasarlamayı kolaylaştırır. Pratik, anlaşılır ve yaratıcı öneriler sunarak öğretmenin hedeflerine ulaşmasını destekler. Ayrıca günlük ders planı tasarımına yönelik basit ve esnek öneriler sunar. Örnek Promptlar Kavram Odaklı Etkinlik Tasarımı "7. sınıf Fen Bilimleri dersinde 'Maddenin Halleri' konusunu daha somut ve eğlenceli hale getirmek için hangi etkinlikleri kullanabilirim?" Beceri Geliştirme Çalışmaları "5. sınıf Türkçe dersinde yazma becerilerini geliştirmek için hangi yaratıcı yazma etkinliklerini önerirsiniz?" Tartışma ve Katılımı Teşvik Edici Stratejiler "11. sınıf Sosyal Bilgiler dersinde 'Demokrasi ve İnsan Hakları' konusunu ele alırken öğrencilerin tartışmaya aktif katılımını sağlamak için hangi stratejiler etkili olabilir?"

**Ders:** Ders Seçiniz veya yazınız **Sınıf:** Sınıf Seçiniz veya yazınız **Tema/Ünite:** Tema/Ünite Seçiniz veya yazı

**Haftalık Ders Saati:** Saat giriniz **Üniteye Ayrılan Ders Saati:** Saat giriniz

Baslat

Ask anything. Type @ / # !

# Your AI Usage Report This Week

**AI Teaching Assistant team** Sun, Feb 9, 2025

**Hello Teacher,**

Here is some data on your use of the AI tool:

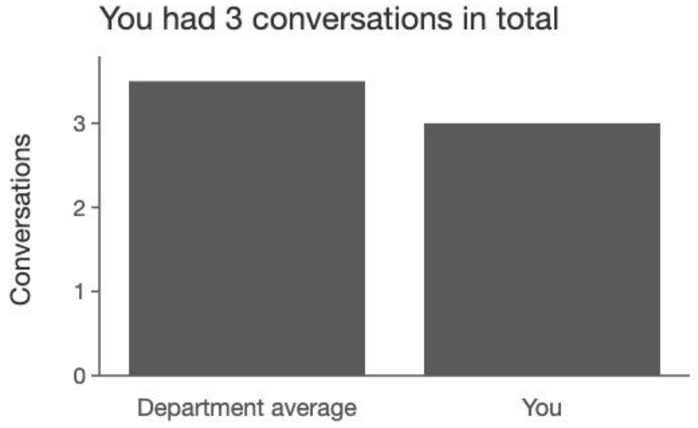
- You sent **8** messages.
- You had **3** conversations.
- You used the AI for about **1** minute in total.

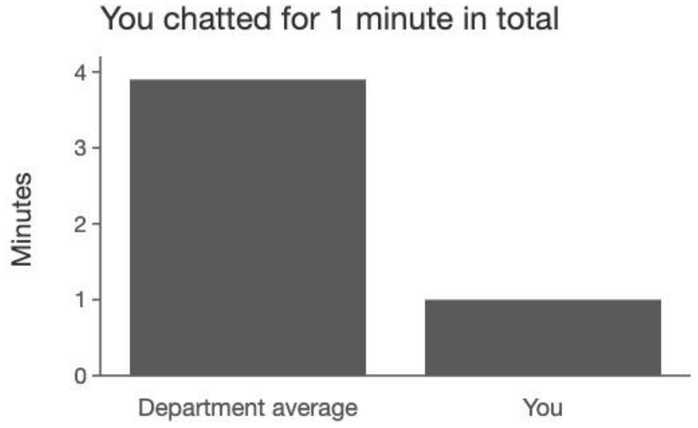
You've made a great start toward exploring more! Wishing you a successful new week.

Feel free to reach out with any questions.

Best wishes



# Timeline

August–December 2024

## Pre-period

Baseline standardized test scores collected

January 2025

## Training

Teachers receive AI training and complete questionnaires

February–May 2025

## Intervention Period

Treated teachers use AI tool
Control teachers teach as usual

Standardized test scores collected

May 2025

## Endline

Teachers and students complete questionnaires

Fall Semester 2024

Winter Break

Spring Semester 2025

# Five questions about what happens when teachers use AI

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# Most teachers did adopt the AI tool

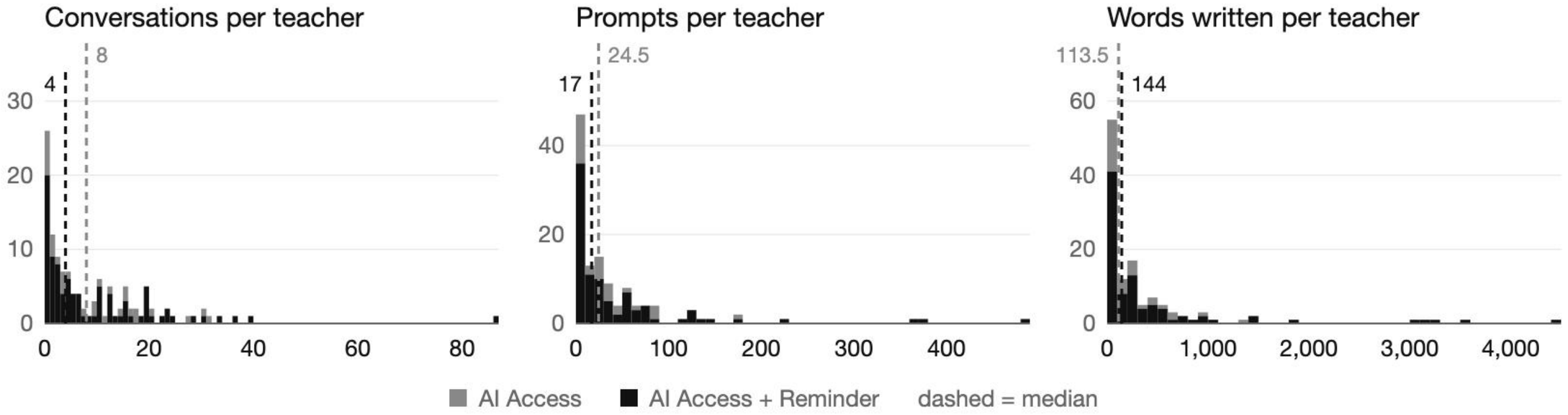
80%

AI Access (24 of 30)

78%

AI Access + Reminder (70 of 90)

Difference: p > .99

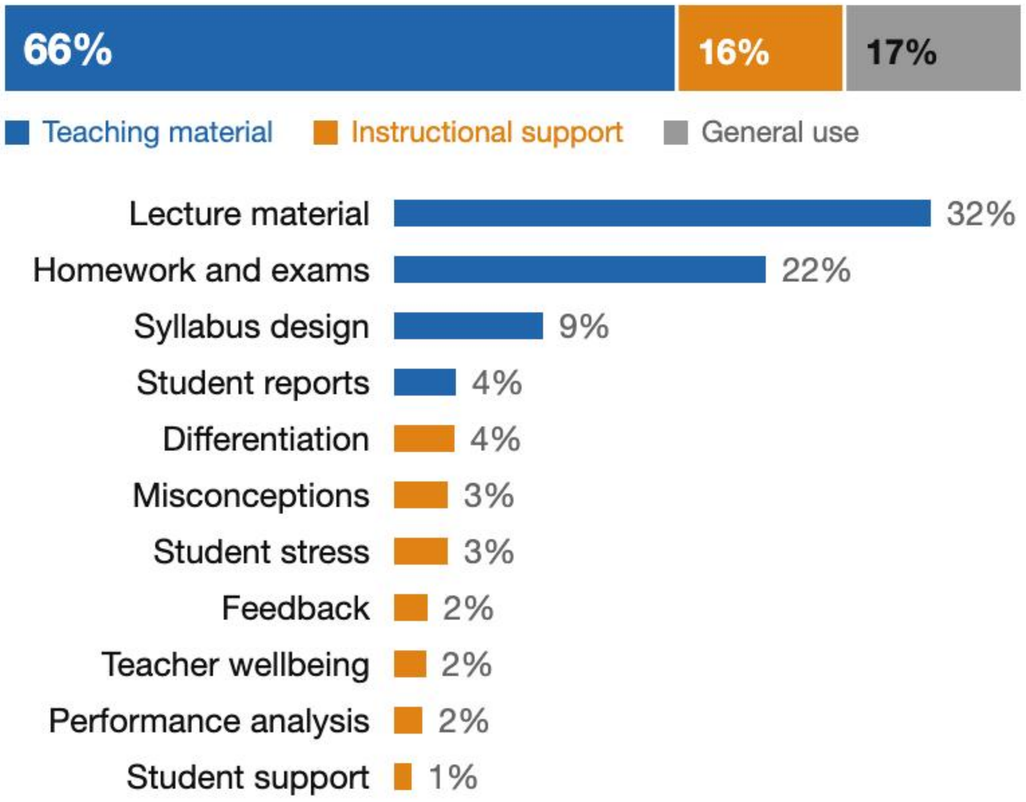


# Five questions about what happens when teachers use AI

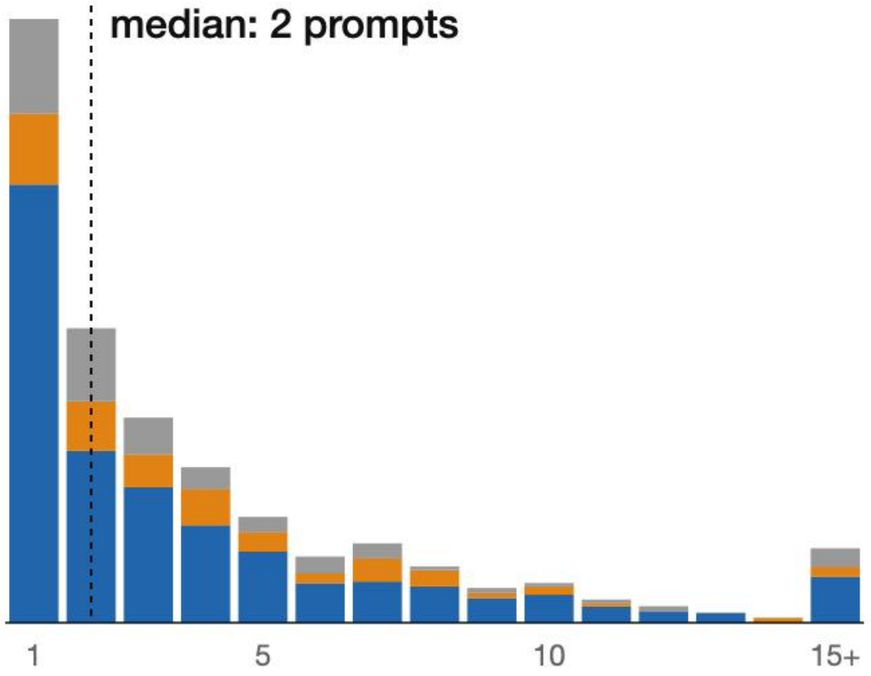
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# Teachers used the AI tool in different ways

1,070 conversations from 94 teachers




Teacher prompts per conversation



# Teachers delegated: they asked for finished materials and rarely iterated

66%

of conversations asked for teaching material: lectures, homework, exams, syllabi

16%

asked for instructional support: differentiation, misconceptions, feedback

2

teacher prompts in the median conversation

# A chemistry teacher accepted the first lesson plan the tool offered

**Teaching material**

Lesson planning: a chemistry teacher, high school. From the tool logs, translated

Make an inference about how scientific knowledge changes through atomic theories.
Course: [Chemistry] Grade: [9th Grade]

Here is a concise plan:
1. Objective: explain how atomic theories changed.
2. Activity: timeline discussion.
3. Assessment: short-answer quiz.
[…]

Thank you, that's suitable.

You're welcome.

# A department head asked the tool how to check on teachers' wellbeing

**Instructional support**

Wellbeing planning: a mathematics department head, high school. From the tool logs, translated

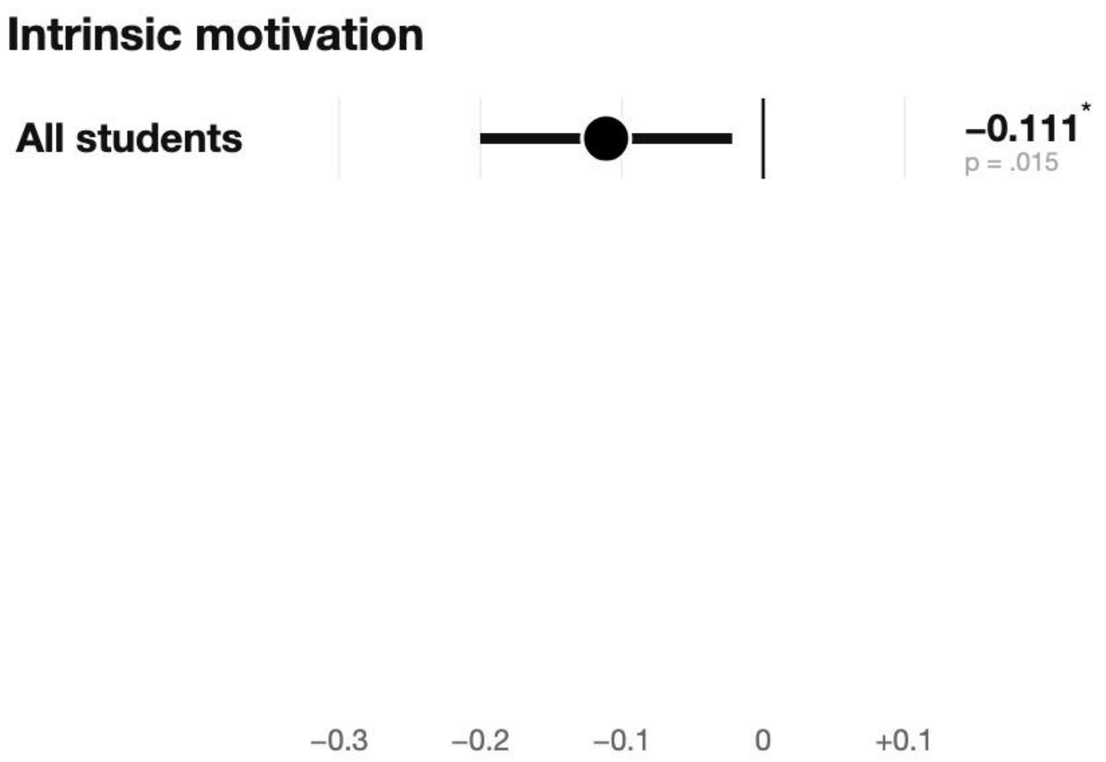
As a department head, how can I identify our teachers' state of wellbeing?

Of course. In high-school mathematics, stress may come from economic difficulties, achievement pressure, and limited resources.
Use anonymous pulse surveys, brief check-ins, and workload review. Then discuss supports such as shared planning time, peer mentoring, clearer priorities, and realistic self-care strategies. […]

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# AI lowered students' intrinsic motivation

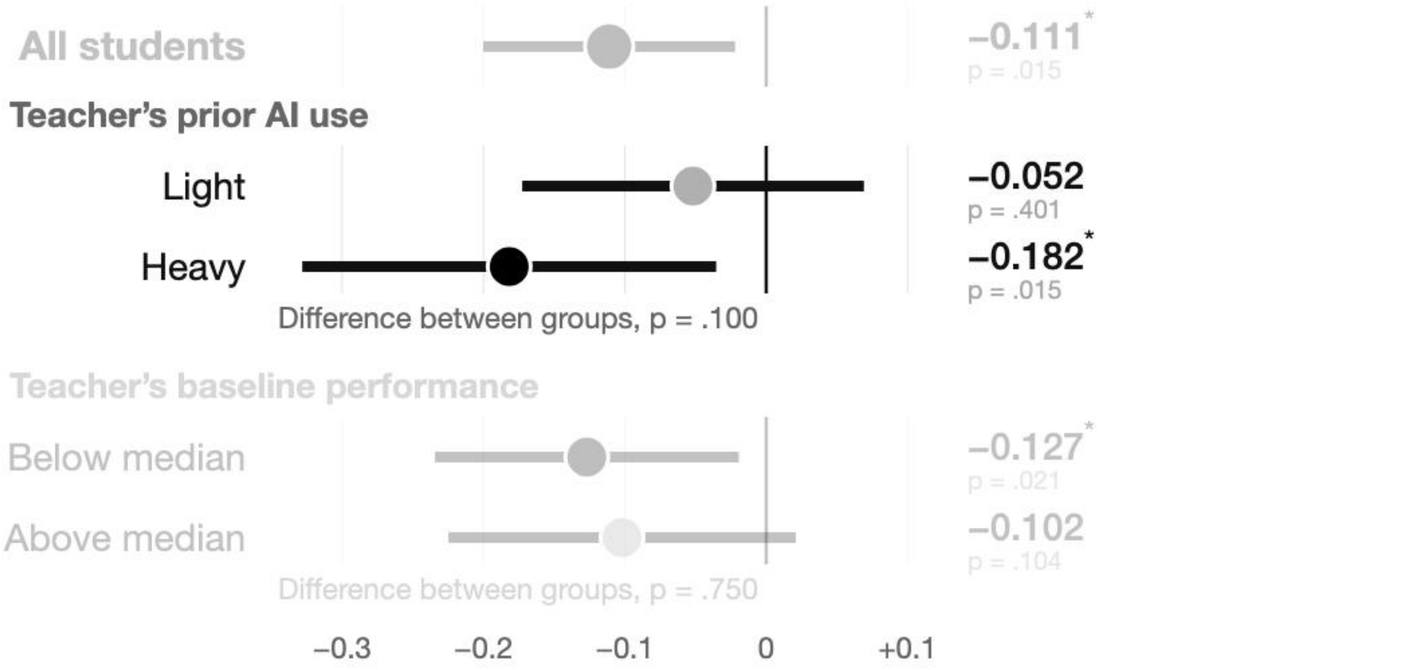



Details

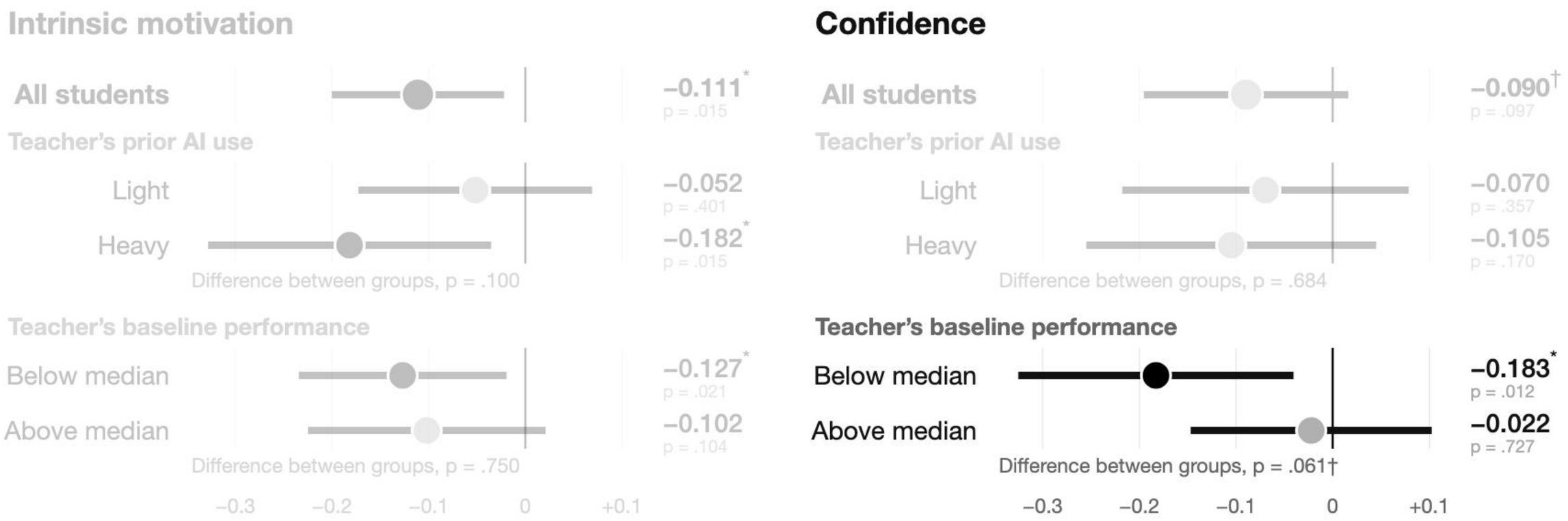
Fredricks et al. 2004

# …especially for teachers higher in baseline AI use

## Intrinsic motivation



# Confidence fell less



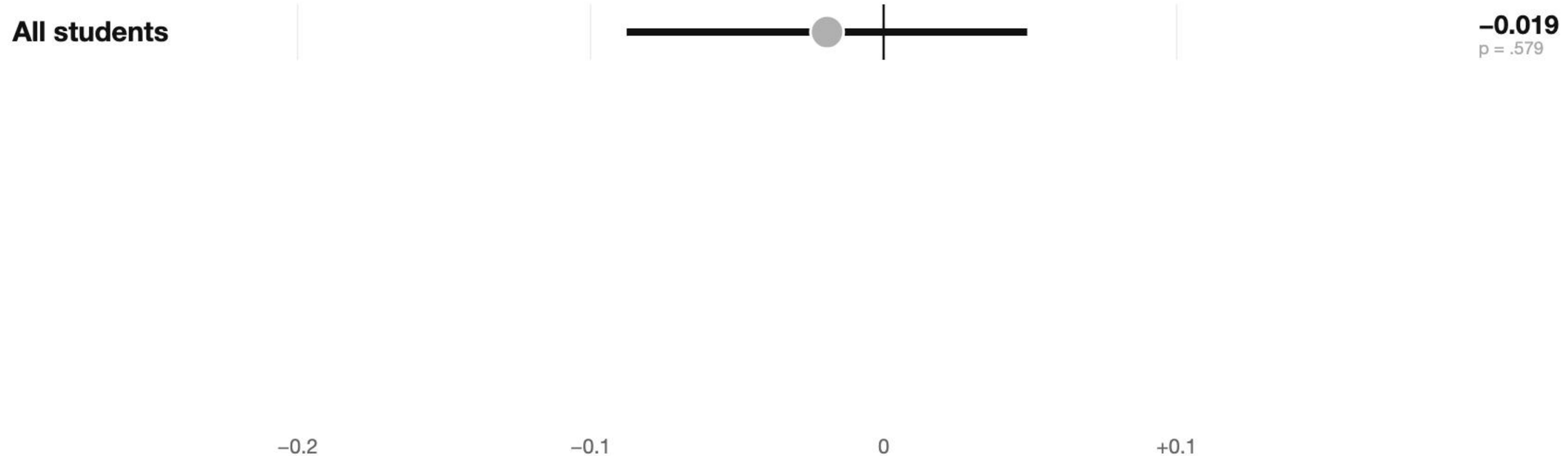

Standard deviations. Bars are 95% CIs

# Five questions about what happens when teachers use AI

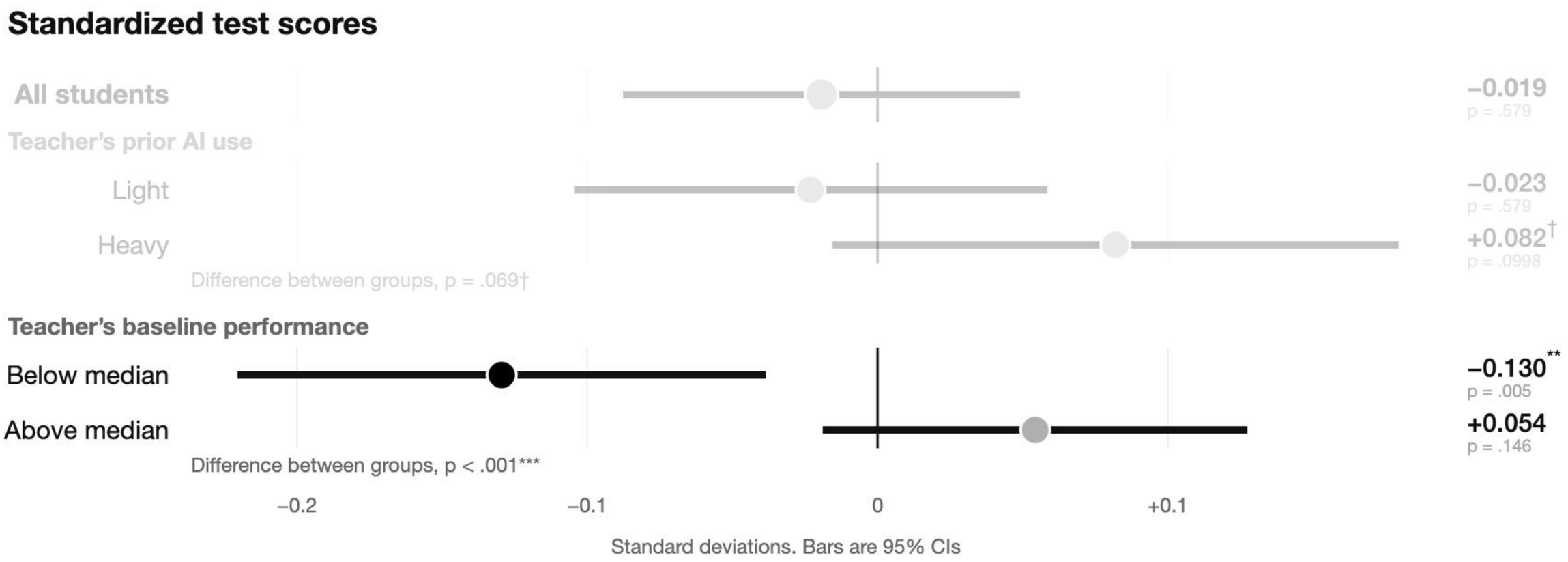
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# On average, AI did not move standardized test scores

## Standardized test scores



# …but it did harm students of lower-performing teachers



# Why was AI harmful?

## Lost voice

AI-generated materials may be perceived as inauthentic.

Doshi & Hauser 2024
Moon et al. 2026

## Skill substitution

Weaker teachers are less able or willing to review, rehearse, internalize and modify AI output.

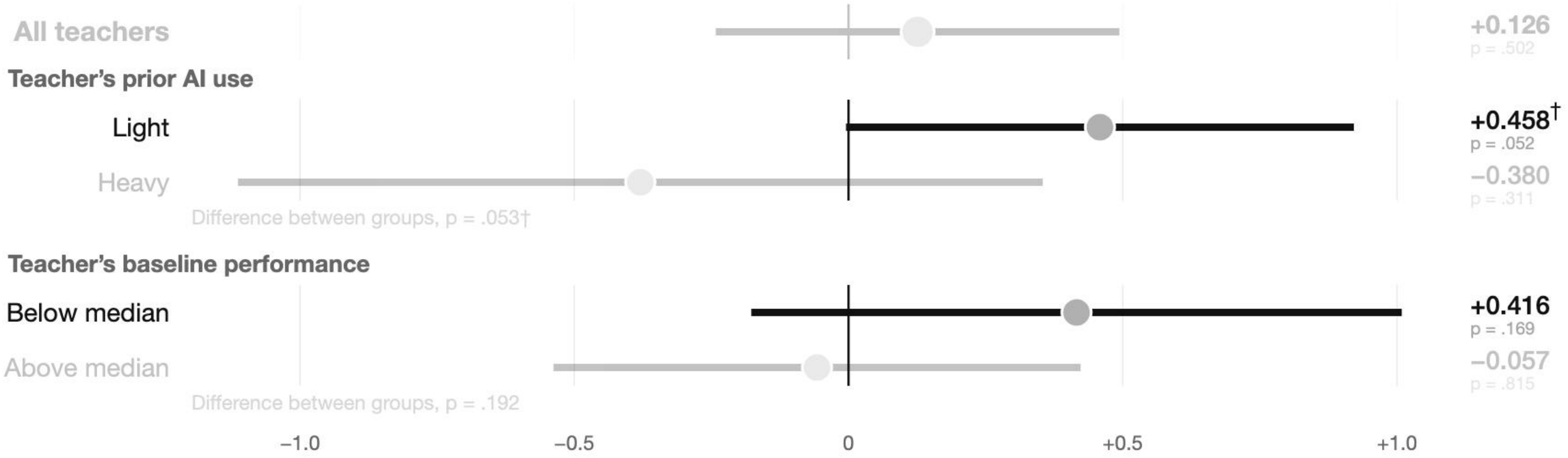
Otis et al. 2026
Dell'Acqua 2022

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# Teachers did not see the cost: a principal–agent problem

## Teacher beliefs about AI




Standard deviations. Bars are 95% CIs

"What overall impact do you believe AI tools have on students' learning outcomes?"

Details

# Limitations




## Duration

We observe one semester.



## Tool specificity

We test one AI tool design.



## Contamination

Control teachers could use free AI tools, so our estimates are conservative.

# We cannot assume AI helps in schools without more careful design

## Optimists

We are too cautious and leave gains on the table.

Gans 2025

## Pessimists

Rapid adoption outruns our understanding of the harms.

Bick et al. 2026

Schools should scale AI for teachers with care.

# Summary

1. In 14 Turkish schools, we **randomly assigned** 193 teachers to get a GPT-4o teaching assistant or teach as usual for one semester.
2. 78% used it, mostly to produce teaching material, with little back-and-forth.
3. Students' intrinsic motivation fell (–0.11 SD), most under teachers who already used AI heavily.
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# Thank you

Benjamin Lira, blira@upenn.edu

Paper

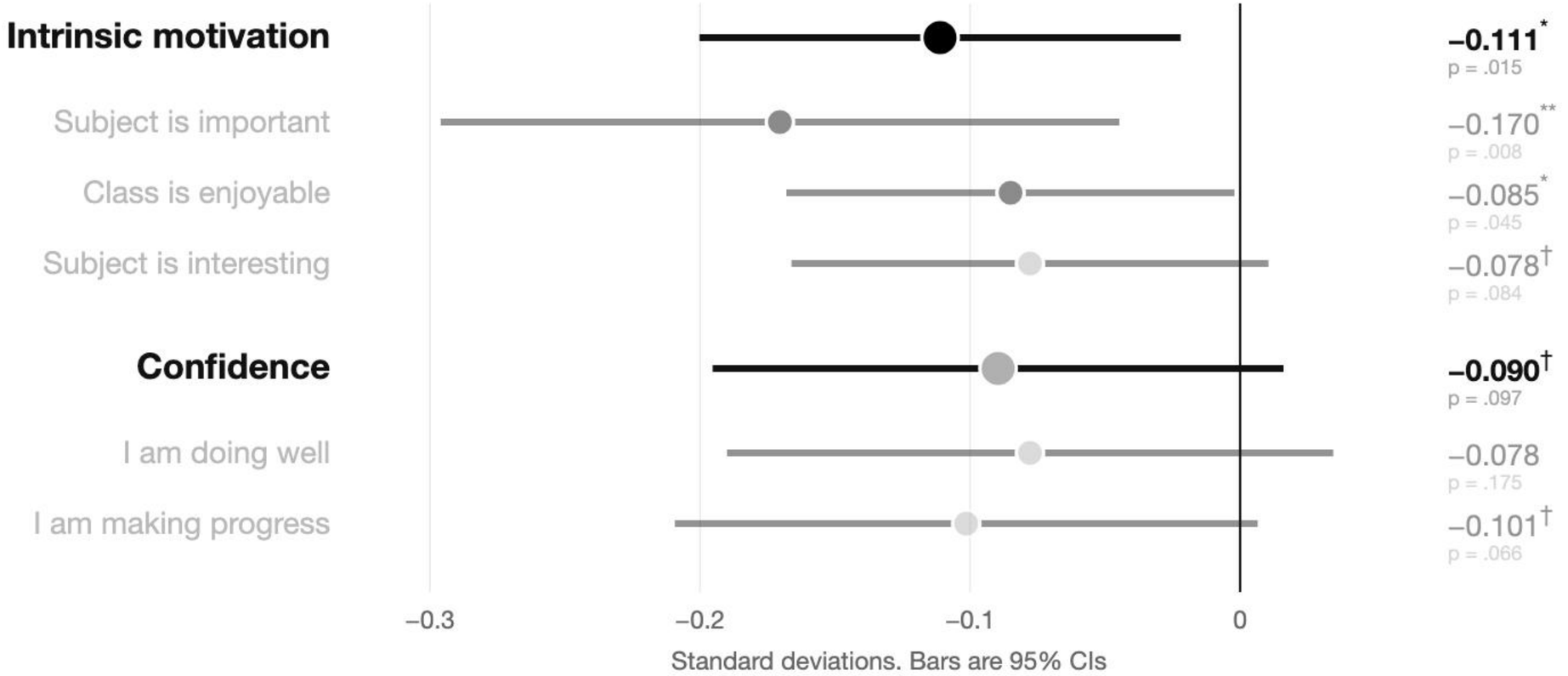
Code and data

# The pattern survives the obvious checks

| | |
|---|---|
| **Attrition** | Response rates did not differ by condition. Lee bounds stay negative. |
| **Within-school variation** | Every school had AI and control teachers. |
| **Fixed effects** | Still negative, though smaller, with department, school and student fixed effects. |
| **Single items** | No one survey item drives the result. |
| **Arms and dosage** | Same pattern for each AI arm and for AI-teacher share. |

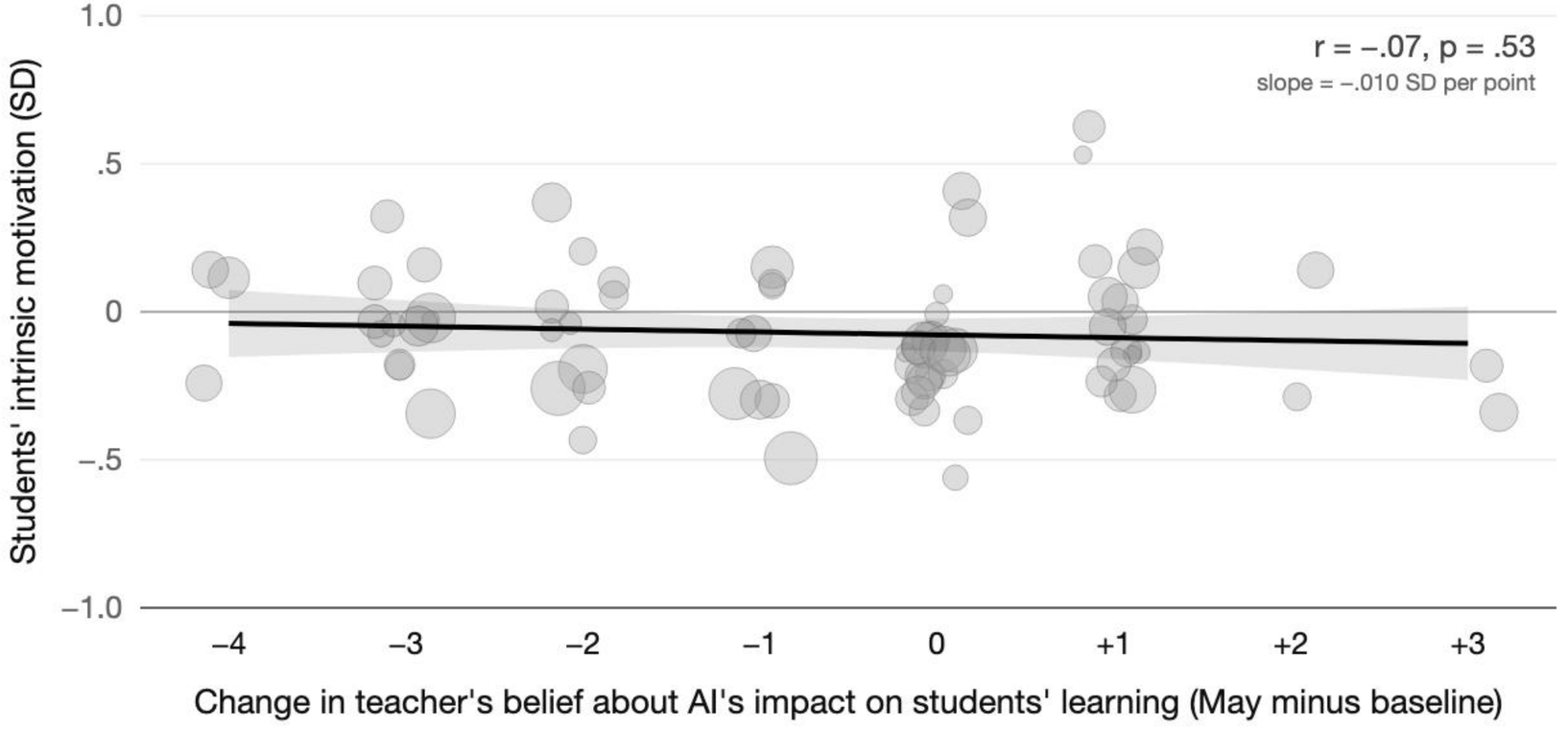
Lee (2009)

# No single survey item drives the result




Back

# Teachers with less intrinsically motivated students did not grow more skeptical of AI

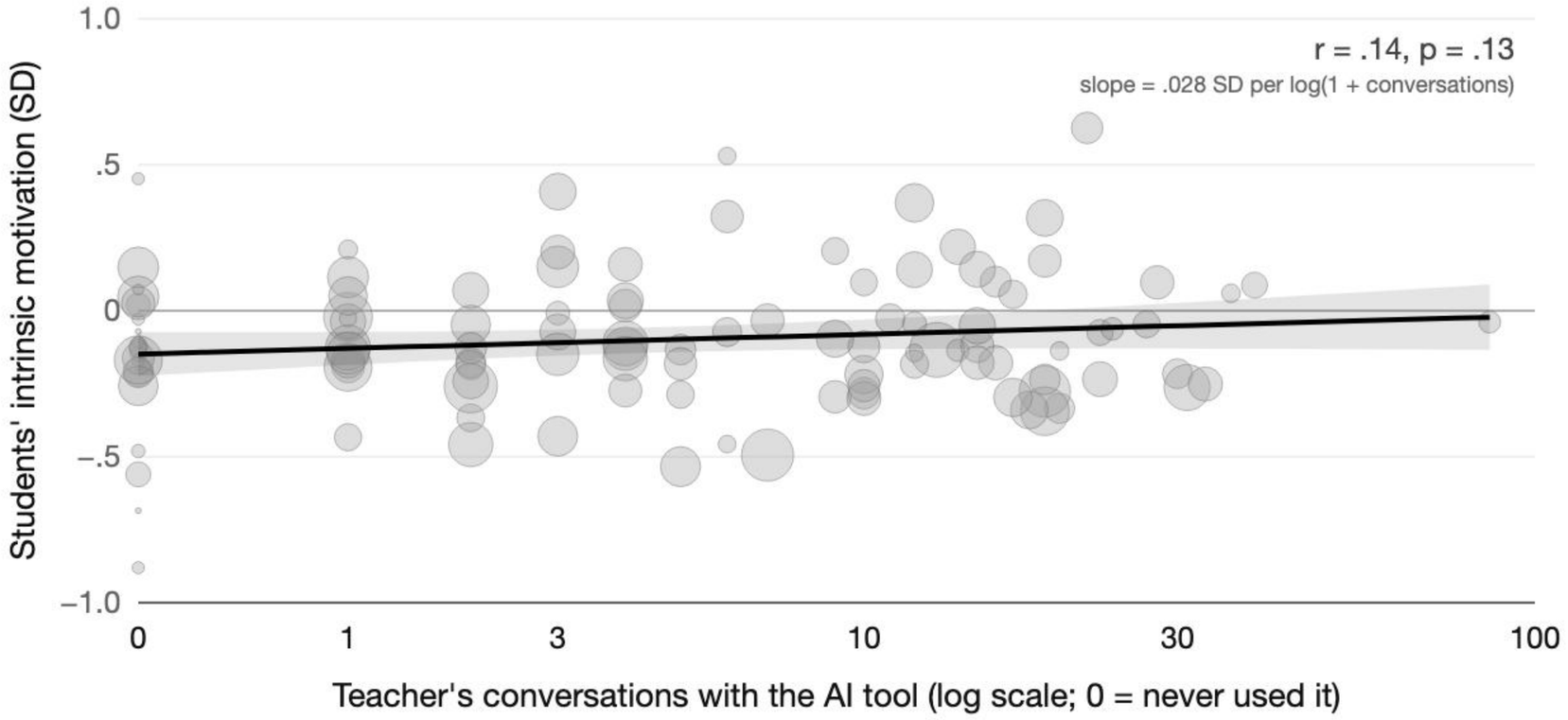



Descriptive; usage and beliefs were not randomized. N = 77 treated teachers. Dot area = number of surveyed students.



Back

# Students' intrinsic motivation did not fall with heavier AI use




Descriptive; usage and beliefs were not randomized. N = 114 treated teachers. Dot area = number of surveyed students.

# Estimation

$$Y_{ic} = \alpha + \boldsymbol{\beta} \cdot \mathrm{AI}_d + \gamma \cdot Y_{ic}^{\mathrm{pre}} + \varepsilon_{ic}$$

*i* student (or teacher), *c* course, *d* school–department (unit of randomization)

**β**: effect of giving teachers AI (intent-to-treat, both AI arms pooled)

γ: baseline control for test scores and teacher beliefs (students had no pre-survey)

Standard errors clustered by school–department

$$Y_{ic} = \alpha + \beta_1 \cdot \mathrm{AI}_d + \beta_2 \cdot M + \boldsymbol{\beta_3} \cdot (\mathrm{AI}_d \times M) + \gamma \cdot Y_{ic}^{\mathrm{pre}} + \varepsilon_{ic}$$

**$\beta_3$**: does the effect differ by the teacher's prior AI use or baseline performance?
*M* = 1 if the teacher is above the median



Back

# The two AI arms look alike, so we pool them

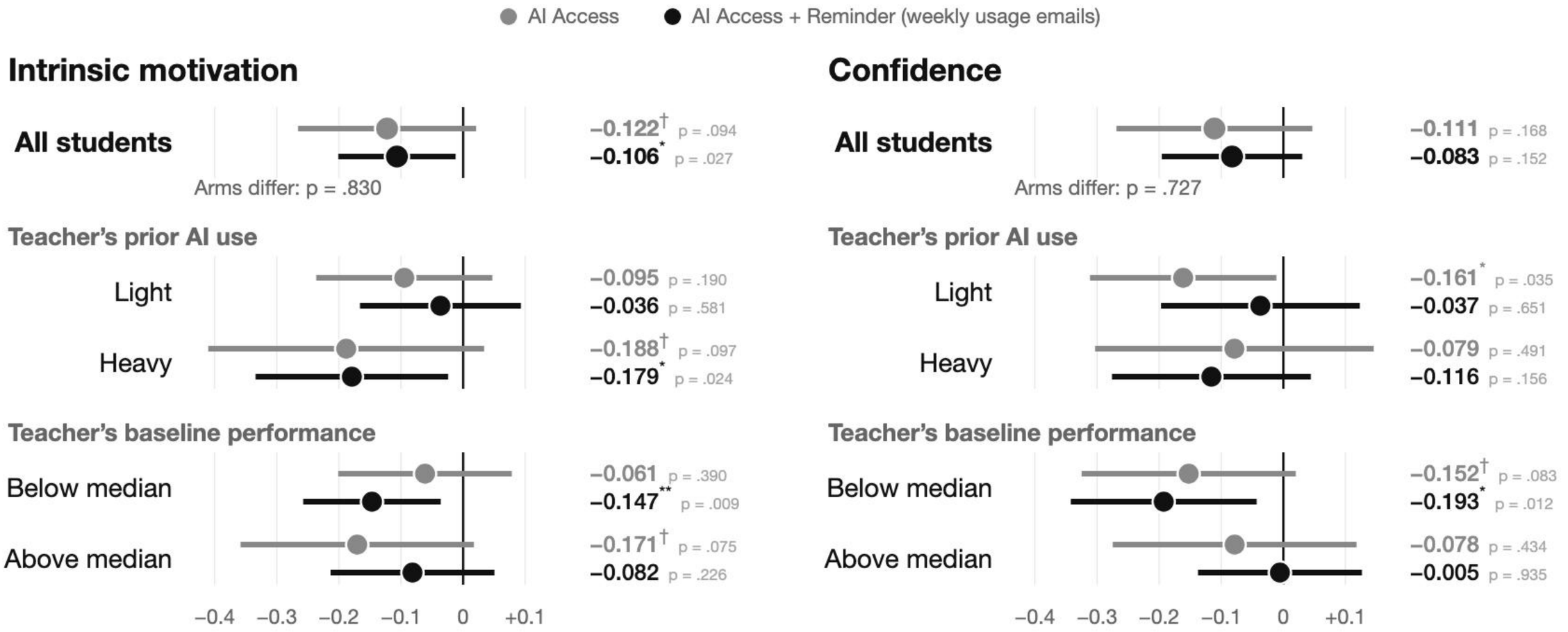
| Outcome | AI Access | AI Access + Reminder | Arms differ (p) |
|---|---|---|---|
| **Adoption** Treated teachers only; no control comparison | | | |
| Used the tool at least once | 80% 24/30 | 78% 70/90 | **1.00** |
| Conversations per teacher | median 8 | median 4 | **.55** |
| **Usage** Treated teachers only; no control comparison | | | |
| Share teaching material | 67% 185/278 | 66% 522/792 | **.93** |
| Prompts per conversation | median 2 | median 3 | **.57** [a] |
| **Intrinsic motivation and confidence** | | | |
| Intrinsic motivation | –0.12† p = .09 | –0.11* p = .03 | **.83** |
| Confidence | –0.11 p = .17 | –0.08 p = .15 | **.73** |
| **Standardized test scores** | –0.01 p = .81 | –0.02 p = .54 | **.73** |
| **Teacher beliefs** | +0.22 p = .38 | +0.09 p = .66 | **.57** |

[a] Means differ: 3.4 vs 5.1 prompts per conversation (p = .02, clustered by teacher).
Teachers: AI Access 30, AI Access + Reminder 90 (analytic sample; 23 and 54 answered the belief survey).
Effects in SD vs control, SEs clustered by school–department.



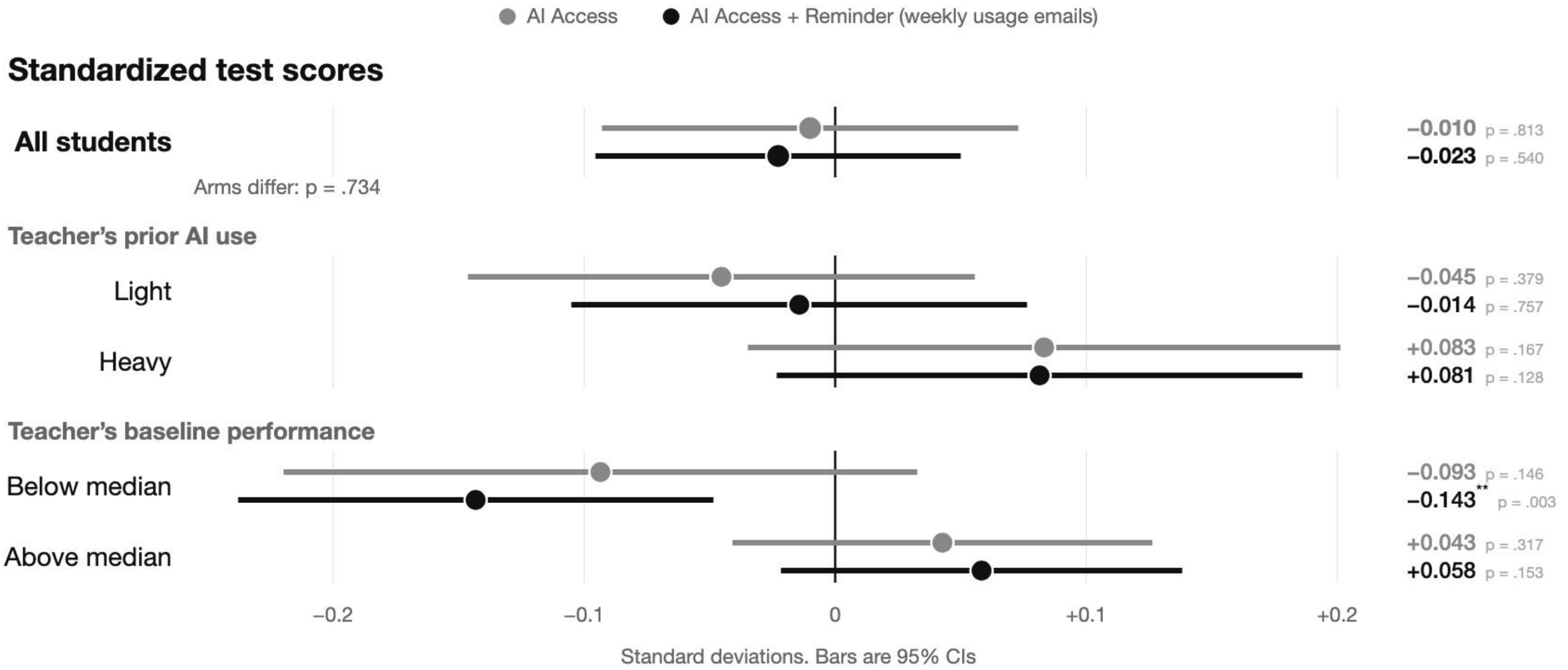
Model Back

# By treatment arm: intrinsic motivation and confidence




Standard deviations. Bars are 95% CIs

# By treatment arm: standardized test scores



# By treatment arm: teacher beliefs about AI

